

# MacSync LCS X1 LoRaWAN® Capacitive Diesel Fuel Level Sensor Datasheet

## Product Overview

**MacSync LCS X1** is an industrial-grade **LoRaWAN® capacitive dip stick diesel fuel level sensor** engineered for accurate real-time fuel-level monitoring across generator fuel tanks, telecom tower diesel reservoirs, industrial fuel storage systems, fleet fuel infrastructure, mining equipment, and remote industrial telemetry deployments.

The device utilizes advanced capacitive dip stick sensing technology to continuously measure diesel fuel levels with high precision, long-term stability, and reliable industrial-grade performance in demanding operating environments. The sensor is designed specifically for diesel tank installations where accurate fuel analytics, low maintenance, and dependable wireless telemetry are critical for operational continuity.

Designed for industrial IoT fuel telemetry applications, the MacSync LCS X1 wirelessly transmits real-time fuel-level analytics using ultra-long-range LoRaWAN® communication for centralized fuel monitoring, consumption analysis, refill tracking, and predictive operational intelligence.

Its rugged industrial enclosure, ultra-low-power electronics, and scalable wireless deployment architecture enable efficient fuel monitoring across DG backup systems, telecom infrastructure, smart utility installations, logistics fleets, industrial facilities, mining environments, and distributed energy systems without requiring complex wired infrastructure.

The sensor is ideal for diesel tank dip stick monitoring, generator fuel analytics, telecom tower fuel management, fleet fuel telemetry, fuel theft detection, industrial fuel storage monitoring, mining equipment fuel analytics, remote diesel telemetry, and smart utility fuel management applications.

MacSync LCS X1 supports BLE-based configuration through the Macnman Maya App, secure OTA configuration, threshold-based fuel alerts, refill event detection, event-driven reporting, and seamless integration with LoRaWAN® gateways, MQTT platforms, SCADA systems, industrial IoT dashboards, and cloud-based fuel analytics platforms.

Whether deployed in industrial plants, telecom infrastructure, mining operations, transportation fleets, backup power systems, utility installations, or remote diesel monitoring environments, the **MacSync LCS X1 LoRaWAN® capacitive dip stick fuel-level sensor** transforms fuel telemetry into actionable operational intelligence for optimized fuel management and predictive monitoring.



## Key Features

### LoRaWAN® Capacitive Fuel Level Sensor

Industrial wireless capacitive fuel-level sensor engineered for diesel tanks, generator fuel systems, industrial fuel reservoirs, telecom infrastructure, and remote fuel telemetry applications.

### Long-Range LoRaWAN® Connectivity

Ultra-long-range wireless telemetry for distributed diesel fuel monitoring across industrial facilities, telecom towers, mining sites, and remote infrastructure.

### Fuel Theft & Consumption Analytics

Monitor fuel consumption patterns, unauthorized fuel removal, refill events, and operational fuel usage analytics in real time.

### Capacitive Dip Stick Fuel Sensor

Industrial capacitive dip stick fuel-level sensing technology engineered for accurate diesel-level measurement inside generator tanks, telecom tower fuel reservoirs, fleet fuel systems, and industrial storage tanks.

### Generator & Diesel Tank Monitoring

Ideal for DG backup systems, telecom tower fuel tanks, industrial fuel reservoirs, and distributed diesel monitoring infrastructure.

### Battery & External Power Variants

Available in ultra-low-power battery-operated models and continuous DC-powered industrial deployment configurations.

---

### Threshold-Based Fuel-Level Alerts

Intelligent event-triggered reporting for low-fuel alerts, refill notifications, overflow conditions, and abnormal fuel consumption detection.

---

### BLE Configuration via Maya App

Simplified local configuration, diagnostics, firmware management, and deployment commissioning using the Macnman Maya Android App.

---

## Applications

---

### Diesel Tank Monitoring

Real-time diesel fuel-level monitoring for industrial fuel tanks, storage reservoirs, backup generators, and distributed fuel infrastructure.

---

### Telecom Tower Fuel Analytics

Wireless fuel-level telemetry for telecom tower diesel tanks deployed in remote and distributed infrastructure environments.

---

### Fuel Theft Detection

Detect unauthorized fuel removal, abnormal fuel-level drops, and suspicious diesel consumption patterns in real time.

---

### Smart Utility Fuel Monitoring

Intelligent fuel-level analytics for smart utility systems, energy infrastructure, and remote power backup installations.

---

### Mining & Heavy Equipment Fuel Analytics

Monitor fuel-level variations and diesel consumption across mining equipment, construction machinery, and industrial vehicles.

---

### Capacitive Fuel-Level Sensing

---

Sensor Type

Capacitive Dip Stick Fuel-Level Sensor

---

Level Range

100 mm to 1400 mm (OEM Customizable)

---

---

### Industrial IP65/IP67 Rugged Enclosure

Rugged industrial housing engineered for outdoor deployments, vibration-prone environments, industrial sites, and harsh fuel monitoring installations.

---

### Compatible with Major LoRaWAN® Platforms

Seamless integration with ChirpStack, TTN, MQTT platforms, SCADA systems, Orbiwise, LORIoT, and private industrial LoRaWAN® networks.

---

---

### Generator Fuel Monitoring

Monitor fuel availability, refill cycles, low-fuel alerts, and generator fuel analytics across DG backup systems.

---

### Fleet Fuel Management

Track diesel usage, refill patterns, fuel consumption, and operational analytics across transportation and logistics fleets.

---

### Industrial Fuel Reservoir Monitoring

Industrial fuel telemetry for factories, utility systems, mining sites, and large-scale diesel storage infrastructure.

---

### Remote Fuel Telemetry

Long-range LoRaWAN® fuel-level monitoring for geographically distributed industrial and utility deployments.

---

|                       |                                                                                                                                                                                                                                 |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Measurement Accuracy  | ±1%                                                                                                                                                                                                                             |
| Resolution            | 0.01 mm                                                                                                                                                                                                                         |
| Operating Temperature | -35°C to +70°C                                                                                                                                                                                                                  |
| Probe Diameter        | Φ16 mm                                                                                                                                                                                                                          |
| Pressure Range        | -0.1 MPa to 0.4 MPa                                                                                                                                                                                                             |
| Material              | Stainless Steel                                                                                                                                                                                                                 |
| Explosion Protection  | Exd II C T5                                                                                                                                                                                                                     |
| Installation Method   | Standard 5-hole flange or thread M20×1.5 (OEM customizable)                                                                                                                                                                     |
| Suitable Applications | Diesel tanks, generator fuel systems, telecom tower fuel monitoring, industrial fuel reservoirs, smart utility infrastructure, fleet fuel monitoring, mining equipment fuel analytics, and remote diesel telemetry applications |

## Wireless Transmission

|                    |                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------|
| Wireless Protocols | LoRaWAN® (v1.0.4), Macnman MacTalk Protocol                                             |
| Antenna            | Internal high-efficiency antenna                                                        |
| Supported Bands    | IN865 / RU864 / EU868 / US915 / AU915                                                   |
| Tx Power           | 23 dBm @ 865 MHz                                                                        |
| Sensitivity        | -137 dBm @ 300 bps                                                                      |
| Activation         | OTAA                                                                                    |
| LoRaWAN® Class     | Class A (Default), Configurable to Class C via Maya                                     |
| Supports Major LNS | Private LNS, ChirpStack, TTN, MQTT, Orbiwise, Netmore, Direct PLC On-Prem Installations |

## Device Management

|                      |                                                |
|----------------------|------------------------------------------------|
| Configuration Method | Via Macnman Maya Android App                   |
| Local Configuration  | BLE-Based Local Configuration                  |
| Remote Configuration | Over-the-Air Configuration (Downlink Commands) |
| Status LED           | X1 Internal Multi-Color LED                    |

## Server Data Reporting Modes

|                        |                                                                    |
|------------------------|--------------------------------------------------------------------|
| Data Reporting Mode    | Periodic / Event-based / Hybrid                                    |
| Transmission Interval  | Independent uplink interval for optimized power consumption        |
| Threshold-Based Alerts | Configurable fuel-level threshold for event-triggered transmission |
| Sampling Interval      | Configurable data acquisition interval (e.g., 1 min to 24 hours)   |

## Battery Performance (BO Series)

|                              |                                                    |
|------------------------------|----------------------------------------------------|
| Battery Type                 | Li-SOCl <sub>2</sub> Cell (19Ah) Industrial grade  |
| Power Consumption            | ~100 µA in sleep mode / ~50 mA during transmission |
| 5-Minute Reporting Interval  | ~2 Years (Estimated)                               |
| 15-Minute Reporting Interval | ~4 Years (Estimated)                               |
| 60-Minute Reporting Interval | ~7+ Years (Estimated)                              |

## Power Performance (PO Series)

|                                        |                                                    |
|----------------------------------------|----------------------------------------------------|
| Input Voltage Range (Optional Variant) | 9-36 V DC (External Supply)                        |
| Power Consumption                      | ~100 µA in sleep mode / ~50 mA during transmission |

## Communication Options

|                  |                                                      |
|------------------|------------------------------------------------------|
| BLE Connectivity | For local configuration and diagnostics via Maya App |
| Cloud Protocols  | MQTT / HTTPS / TCP IP (Gateway Dependent)            |

## Security & Encryption

|                     |                                    |
|---------------------|------------------------------------|
| Data Encryption     | AES-128 LoRaWAN® encryption        |
| Authentication      | OTAA Device Activation             |
| Firmware Protection | Secure boot and signed OTA updates |

## LED Status

|          |                            |
|----------|----------------------------|
| LED Type | 1 Internal Multi-Color LED |
|----------|----------------------------|

|                     |                                                    |
|---------------------|----------------------------------------------------|
| Solid Blue          | Device is connected with Maya App                  |
| Flashing Blue       | OTA (Over-the-Air) update in progress              |
| Slow Blinking Green | Attempting to connect to the LoRaWAN® network      |
| Single Red Blink    | A confirmed parameter of MacSync is being modified |

### Physical Characteristics

|                       |                                       |
|-----------------------|---------------------------------------|
| Operating Temperature | -40°C to +80°C                        |
| Humidity Range        | 0% to 100% RH (non-condensing)        |
| Ingress Protection    | IP65 standard (IP67 optional variant) |
| Housing Material      | Glass Filled Nylon                    |
| Color                 | Matte White                           |
| Weight                | ~175g                                 |
| Mounting Options      | Pole Mounting / Wall Mounting         |

### Certifications & Compliance

|      |                            |
|------|----------------------------|
| WPC  | Compliant                  |
| RoHS | Conforms to RoHS directive |

## MacSync LCS X1 Variants

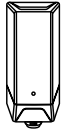
| SKU  | VARIANT NAME        | MEASUREMENT RANGE  | POWER SOURCE    | COMMUNICATION | IDEAL FOR                                                                                                                                                                 |
|------|---------------------|--------------------|-----------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9116 | MacSync-L-HPT-BS-X4 | -10 kPa to +10 kPa | Battery (19 Ah) | LoRaWAN®      | Remote hydro-static liquid level monitoring for soap dispensers, water tanks, diesel tanks, chemical containers, beverage systems, and smart liquid storage applications. |

| SKU  | VARIANT NAME        | MEASUREMENT RANGE  | POWER SOURCE    | COMMUNICATION | IDEAL FOR                                                                                                                              |
|------|---------------------|--------------------|-----------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 9117 | MacSync-L-HPT-PS-X4 | -10 kPa to +10 kPa | 9–36 V DC       | LoRaWAN®      | Continuous hydro-static liquid level monitoring for industrial tanks, utility reservoirs, process vessels, and liquid storage systems. |
| 9118 | MacSync-M-HPT-BS-X4 | -10 kPa to +10 kPa | Battery (19 Ah) | MacTalk       | Battery-operated MacTalk deployments for remote hydro-static liquid level monitoring and industrial utility management.                |
| 9119 | MacSync-M-HPT-PS-X4 | -10 kPa to +10 kPa | 9–36 V DC       | MacTalk       | MacTalk-based hydro-static liquid level monitoring with continuous external power for industrial automation.                           |
| 9120 | MacSync-B-HPT-BS-X4 | -10 kPa to +10 kPa | Battery (19 Ah) | BLE           | Local hydro-static liquid level monitoring, commissioning, calibration, and diagnostics using BLE connectivity.                        |

## What's in the Box?

Make sure the package contains the following items before installation and deployment of the MacSync LCS X1 LoRaWAN® Capacitive Fuel-Level Sensor.

|                                                              |                         |                   |
|--------------------------------------------------------------|-------------------------|-------------------|
| ✓ MacSync LCS X1<br>LoRaWAN® Capacitive<br>Fuel-Level Sensor | ✓ Mounting Clip         | ✓ User Manual     |
| ✓ Warranty Card                                              | ✓ IP68 Cable Extender * | ✓ Mounting Screws |
| ✓ Power Adapter *                                            |                         |                   |



MacSync



Mounting  
Clip



User manual &  
Warranty Card



Junction  
Box\*



Mounting  
Screws



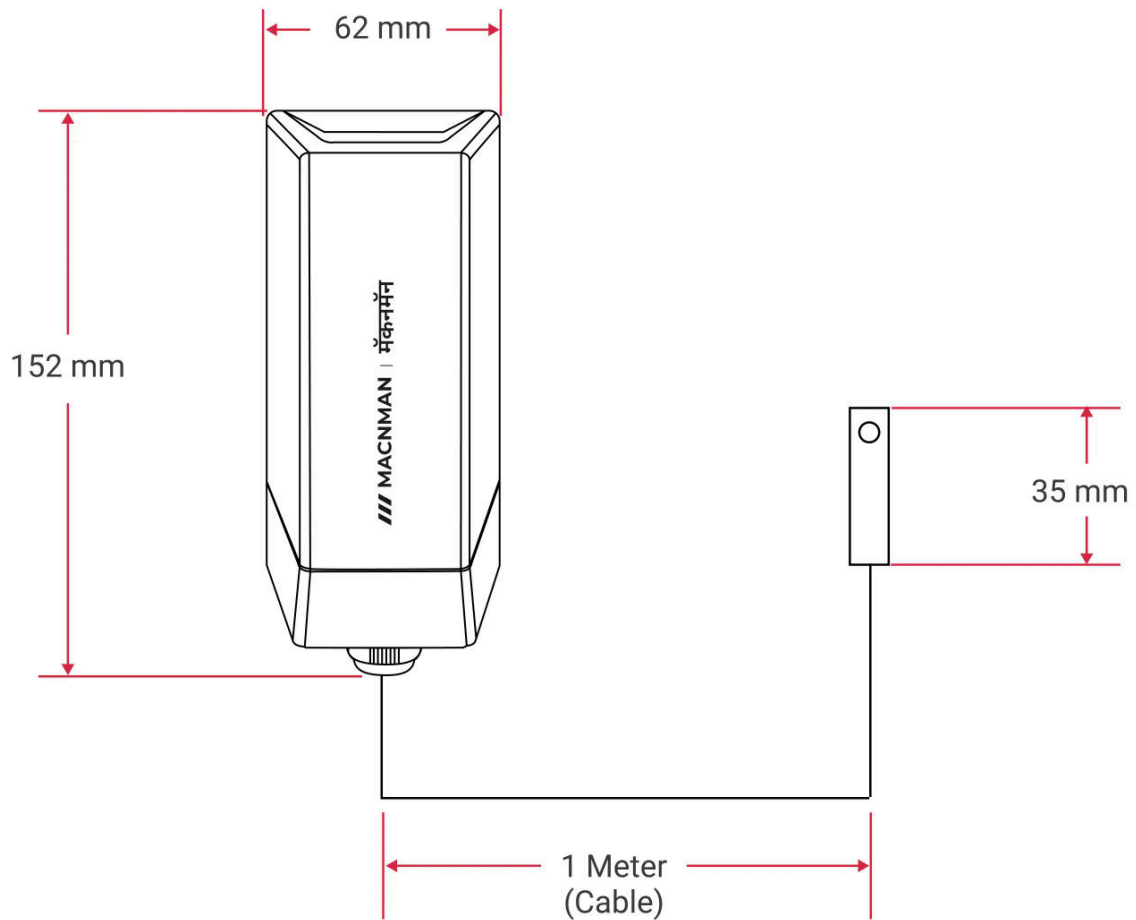
Power  
Adapters\*

**Note**

Accessories marked with (\*) may vary depending on deployment requirements and ordering configuration.

## MacSync LCS X1 Dimensions

**MacSync LCS X1 BO** features a compact industrial enclosure designed for rugged field deployments with overall dimensions **152 mm X 62 mm X 43 mm (Without Sensor Probe)** optimized for industrial diesel fuel tank monitoring and LoRaWAN® fuel telemetry installations.



## Frequently Asked Questions

### What is the MacSync LCS X1 used for?

MacSync LCS X1 is an industrial LoRaWAN capacitive diesel fuel level sensor for generator tanks, fuel storage, fleet and telecom tower fuel monitoring. Typical applications include Diesel Tank Monitoring, Generator Fuel Monitoring, Telecom Tower Fuel Analytics, Fleet Fuel Management, Fuel Theft Detection and Industrial Fuel Reservoir Monitoring.

### What does the MacSync LCS X1 measure, and how accurately?

Measurement specifications of the MacSync LCS X1:

- Capacitive Fuel-Level: range 100 mm to 1400 mm (OEM Customizable), accuracy  $\pm 1\%$ , resolution 0.01 mm.

## Which LoRaWAN frequency bands and network servers does the MacSync LCS X1 support?

The MacSync LCS X1 supports LoRaWAN® (v1.0.4) and the Macnman MacTalk Protocol on the IN865 / RU864 / EU868 / US915 / AU915 frequency bands. It joins the network by OTAA activation. It runs as Class A by default and can be set to Class C in the Maya app. Supported network servers: Private LNS, ChirpStack, TTN, MQTT, Orbiwise, Netmore, Direct PLC On-Prem Installations.

## How long does the MacSync LCS X1 battery last?

The battery-operated MacSync LCS X1 runs on a 19 Ah industrial-grade Li-SOCl<sub>2</sub> cell. Estimated battery life is ~2 years at a 5-minute reporting interval, ~4 years at a 15-minute reporting interval and ~7+ years at a 60-minute reporting interval. A power-operated version runs from 9-36 V DC (External Supply).

## What is the IP rating and operating temperature of the MacSync LCS X1?

The MacSync LCS X1 has an ingress protection rating of IP65 standard (IP67 optional variant) and an operating temperature range of -40°C to +80°C. Operating humidity: 0% to 100% RH (non-condensing). Mounting options: Pole Mounting / Wall Mounting.

## How is the MacSync LCS X1 configured?

The MacSync LCS X1 is configured with the Macnman Maya Android app over Bluetooth Low Energy (BLE). Settings can also be changed remotely with over-the-air downlink commands.

## What is included in the box with the MacSync LCS X1?

The box contains: MacSync LCS X1 LoRaWAN® Capacitive Fuel-Level Sensor, Mounting Clip, User Manual, Warranty Card and Mounting Screws. IP68 Cable Extender and Power Adapter are optional and may need to be purchased separately.

## Related Products and Resources

- Setup guide: [Common LoRaWAN sensor settings in the Maya app](#)
- Compatible LoRaWAN gateways: [MacLink Gen-2 industrial outdoor LoRaWAN gateway](#) and [Setu LX Series indoor LoRaWAN gateway](#)
- Related datasheets: [MacRay LUS X1 LoRaWAN ultrasonic level sensor](#), [MacRay LToF LoRaWAN Time-of-Flight sensor](#)
- Product page: [MacSync LCS X1 on macnman.com](#)
- Help: [Warranty terms and conditions](#) and [contact Macnman support](#)